Screenshots:

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ make
gcc -O2 -Wall -Wextra -o engine engine.c -lpthread
engine.c: In function ‘child_fn’:
engine.c:540:5: warning: ignoring return value of ‘chdir’ declared with attribute ‘warn_unused_result
’ [-Wunused-result]
  540 |     chdir("/");
      |     ^~~~~~~~~~
At top level:
engine.c:282:13: warning: ‘bounded_buffer_begin_shutdown’ defined but not used [-Wunused-function]
  282 | static void bounded_buffer_begin_shutdown(bounded_buffer_t *buffer)
      |             ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~
engine.c:275:13: warning: ‘bounded_buffer_destroy’ defined but not used [-Wunused-function]
  275 | static void bounded_buffer_destroy(bounded_buffer_t *buffer)
      |             ^~~~~~~~~~~~~~~~~~~~~~
gcc -O2 -Wall -static -o memory_hog memory_hog.c
gcc -O2 -Wall -static -o cpu_hog cpu_hog.c
gcc -O2 -Wall -static -o io_pulse io_pulse.c
make -C /lib/modules/6.8.0-47-generic/build M=/home/vishal/OS-Jackfruit/boilerplate modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-47-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04) 12.3.0
  You are using:           gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0
  CC [M]  /home/vishal/OS-Jackfruit/boilerplate/monitor.o
  MODPOST /home/vishal/OS-Jackfruit/boilerplate/Module.symvers
  CC [M]  /home/vishal/OS-Jackfruit/boilerplate/monitor.mod.o
  LD [M]  /home/vishal/OS-Jackfruit/boilerplate/monitor.ko
  BTF [M] /home/vishal/OS-Jackfruit/boilerplate/monitor.ko
Skipping BTF generation for /home/vishal/OS-Jackfruit/boilerplate/monitor.ko due to unavailability of
 vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-47-generic'
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo insmod monitor.ko
ls -l /dev/container_monitor
crw------- 1 root root 510, 0 Apr 23 12:42 /dev/container_monitor
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

1

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ cd ~/OS-Jackfruit/boilerplate
sudo ./engine start alpha ~/OS-Jackfruit/rootfs-alpha /bin/sh --soft-mib 48 --hard-mib 80
sudo ./engine start beta ~/OS-Jackfruit/rootfs-beta /bin/sh --soft-mib 64 --hard-mib 96
[sudo] password for vishal:
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ~/OS-Jackfruit/rootfs-base
Supervisor started with base rootfs: /home/vishal/OS-Jackfruit/rootfs-base
Starting container alpha
Started container alpha with PID 53750
Starting container beta
Started container beta with PID 53755
Finished writing logs
Container with PID 53750 exited
Container with PID 53755 exited
Finished writing logs
```

2

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
Expected states include: starting, running, stopped, killed, exited
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
Supervisor started with base rootfs: /home/vishal/OS-Jackfruit/rootfs-base
Starting container alpha
Started container alpha with PID 17066
Starting container beta
Started container beta with PID 17071
ID      PID     STATE
beta    17071   RUNNING
alpha   17066   RUNNING
Container with PID 17066 exited
Finished writing logs
Container with PID 17071 exited
Finished writing logs
```

3

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine logs alpha
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
Finished writing logs
Running container logtest in foreground
Started container logtest with PID 19654
Finished writing logs
Container logtest finished
```

4

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine stop alpha
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
Stopping container alpha
```

5,6

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep container_monitor | tail -30
[ 1636.559720] [container_monitor] PID=17071 RSS=59375616 SOFT=67108864 HARD=100663296
[ 1636.559732] [container_monitor] PID=17066 RSS=59244544 SOFT=50331648 HARD=83886080
[ 1637.583741] [container_monitor] PID=17071 RSS=67764224 SOFT=67108864 HARD=100663296
[ 1637.583764] [container_monitor] SOFT LIMIT container=beta pid=17071 rss=67764224 limit=67108864
[ 1637.583771] [container_monitor] PID=17066 RSS=67633152 SOFT=50331648 HARD=83886080
[ 1638.607769] [container_monitor] PID=17071 RSS=76152832 SOFT=67108864 HARD=100663296
[ 1638.607785] [container_monitor] PID=17066 RSS=76021760 SOFT=50331648 HARD=83886080
[ 1639.631796] [container_monitor] PID=17071 RSS=84541440 SOFT=67108864 HARD=100663296
[ 1639.631808] [container_monitor] PID=17066 RSS=84410368 SOFT=50331648 HARD=83886080
[ 1639.631817] [container_monitor] HARD LIMIT container=alpha pid=17066 rss=84410368 limit=83886080
[ 1640.655812] [container_monitor] PID=17071 RSS=92930048 SOFT=67108864 HARD=100663296
[ 1641.679837] [container_monitor] PID=17071 RSS=101318656 SOFT=67108864 HARD=100663296
[ 1641.679868] [container_monitor] HARD LIMIT container=beta pid=17071 rss=101318656 limit=100663296
[ 1815.591596] [container_monitor] Registering container=logtest pid=19654 soft=41943040 hard=67108864
[ 1815.762748] [container_monitor] PID=19654 RSS=9043968 SOFT=41943040 HARD=67108864
[ 1816.786763] [container_monitor] PID=19654 RSS=17432576 SOFT=41943040 HARD=67108864
[ 1817.810772] [container_monitor] PID=19654 RSS=25821184 SOFT=41943040 HARD=67108864
[ 1818.834784] [container_monitor] PID=19654 RSS=34209792 SOFT=41943040 HARD=67108864
[ 1819.858797] [container_monitor] PID=19654 RSS=42598400 SOFT=41943040 HARD=67108864
[ 1819.858813] [container_monitor] SOFT LIMIT container=logtest pid=19654 rss=42598400 limit=41943040
[ 1820.882813] [container_monitor] PID=19654 RSS=50987008 SOFT=41943040 HARD=67108864
[ 1821.906819] [container_monitor] PID=19654 RSS=59375616 SOFT=41943040 HARD=67108864
[ 1822.930827] [container_monitor] PID=19654 RSS=67764224 SOFT=41943040 HARD=67108864
[ 1822.930853] [container_monitor] HARD LIMIT container=logtest pid=19654 rss=67764224 limit=67108864
[ 2112.200502] [container_monitor] Registering container=memtest pid=23135 soft=10485760 hard=20971520
[ 2112.724715] [container_monitor] PID=23135 RSS=9043968 SOFT=10485760 HARD=20971520
[ 2113.748728] [container_monitor] PID=23135 RSS=17432576 SOFT=10485760 HARD=20971520
[ 2113.748740] [container_monitor] SOFT LIMIT container=memtest pid=23135 rss=17432576 limit=10485760
[ 2114.772720] [container_monitor] PID=23135 RSS=25821184 SOFT=10485760 HARD=20971520
[ 2114.772734] [container_monitor] HARD LIMIT container=memtest pid=23135 rss=25821184 limit=20971520
```

```
Running container memtest in foreground
Started container memtest with PID 23135
Finished writing logs
Container memtest finished
Running container cpu_normal in foreground
Started container cpu_normal with PID 31062
Finished writing logs
Container cpu_normal finished
```

7

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ time sudo ./engine run cpu_normal ~/OS-Jackfruit/rootfs-alpha /cpu_hog
time nice -n 19 sudo ./engine run cpu_low ~/OS-Jackfruit/rootfs-beta /cpu_hog

real    0m0.031s
user    0m0.007s
sys     0m0.019s

real    0m0.035s
user    0m0.009s
sys     0m0.022s
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

```
Running container cpu_normal in foreground
Started container cpu_normal with PID 54470
Finished writing logs
Container cpu_normal finished
Starting container cpu_hi
Started container cpu_hi with PID 54516
Starting container cpu_lo
Started container cpu_lo with PID 54523
ID       PID      STATE
cpu_lo   54523    RUNNING
cpu_hi   54516    RUNNING
cpu_normal        54470    RUNNING
memtest 54084     STOPPED
beta     53755    STOPPED
alpha    53750    STOPPED
Finished writing logs
Container with PID 54523 exited
Container with PID 54516 exited
Finished writing logs
```

8

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo ./engine stop beta
sudo ./engine ps
ps aux | grep engine
sudo rmmod monitor
Expected states include: starting, running, stopped, killed, exited
root       16713  0.0  0.0  23332  5888 pts/1    S+   12:19   0:00 sudo ./engine supervisor /home/vishal
/OS-Jackfruit/rootfs-base
root       16714  0.0  0.0  23332  2484 pts/0    Ss   12:19   0:00 sudo ./engine supervisor /home/vishal
/OS-Jackfruit/rootfs-base
root       16715  0.0  0.0 179976  1768 pts/0    S+   12:19   0:00 ./engine supervisor /home/vishal/OS-J
ackfruit/rootfs-base
vishal     43794  0.0  0.0   9532  2560 pts/2    S+   12:39   0:00 grep --color=auto engine
```

```
vishal@vishal:~/OS-Jackfruit/boilerplate$ sudo rmmod monitor 2>/dev/null
sudo pkill -f engine 2>/dev/null
ps aux | grep engine
vishal     45278  0.0  0.0   9400  2560 pts/2    S+   12:40   0:00 grep --color=auto engine
vishal@vishal:~/OS-Jackfruit/boilerplate$
```

Screenshot 0,1 — Multi-container supervision

The project is built successfully using `make`, the kernel module is loaded with `sudo insmod monitor.ko`, and the `/dev/container_monitor` device is verified. The supervisor is then started and launches both container alpha (PID 53750) and container beta (PID 53755) under a single supervisor process.

Screenshot 2 — Metadata tracking

Output of `sudo ./engine ps` showing tracked container metadata including container ID, PID, and state. Both alpha (PID 17066) and beta (PID 17071) are shown as RUNNING at the time of the ps call, confirming the supervisor is actively tracking both containers.

Screenshot 3 — Bounded-buffer logging

Output of `sudo ./engine logs alpha` shows the logging pipeline operating. The supervisor terminal shows "Finished writing logs" after the logtest container (PID 19654) completes, confirming the producer-consumer bounded buffer pipeline captured and flushed the container's output to the log file.

Screenshot 4 — CLI and IPC

The CLI command `sudo ./engine stop alpha` is issued from the client terminal (top). The supervisor process receives the command over the UNIX domain socket and responds with "Stopping container alpha" (bottom), demonstrating the second IPC mechanism — the control channel between CLI client and supervisor.

Screenshot 5 — Soft-limit warning

`dmesg` output filtered for `container_monitor` shows soft-limit warning events. For example: `SOFT LIMIT container=beta pid=17071 rss=67764224 limit=67108864` — the kernel module detected that beta's RSS exceeded its soft threshold of 64 MiB and logged a warning.

Screenshot 6 — Hard-limit enforcement

The same `dmesg` output shows hard-limit enforcement. `HARD LIMIT container=alpha pid=17066 rss=84410368 limit=83886080` — the kernel module sent SIGKILL to alpha after its RSS exceeded the hard limit of 80 MiB. Similarly for beta: `HARD LIMIT container=beta pid=17071 rss=101318656 limit=100663296`. The memtest container (PID 23135) also shows both soft and hard limit events with a 10 MiB soft / 20 MiB hard configuration.

Screenshot 7 — Scheduling experiment

Two CPU-bound workloads run using `cpu_hog`: one at normal priority (`time sudo ./engine run cpu_normal ... /cpu_hog`) and one at lower priority (`time nice -n 19 sudo ./engine run cpu_low ... /cpu_hog`). The normal priority run completes in 0.031s real time while the lower priority run takes 0.035s real time, showing the effect of the CFS scheduler weighting. The supervisor terminal confirms both containers were tracked and completed.

Screenshot 8 — Clean teardown

After stopping all containers and running `sudo rmmod monitor` and `sudo pkill -f engine`, the final `ps aux | grep engine` shows only the grep process itself — no engine supervisor or container processes remain. This confirms all containers were reaped, logging threads exited cleanly, and no zombie processes remain.